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directional language; (s1dec) [decision, below of=s1, yshift=-0.3cm] Confident
match?;
(s2) [stage, below of=s1dec, yshift=-0.3cm, xshift=-2.5cm, fill=orange!20] Stage 2:
Structured
Extraction; (s2note) [label, right of=s2, xshift=3.0cm] LLM extracts 5 binary features:
claims_superiority,
has_directional_claim,
mentions_no_difference,
has_hedging, cites_evidence;
(s3) [stage, below of=s2, yshift=-0.2cm, fill=green!20] Stage 3: Deterministic
Mapping; (s3note) [label, right of=s3, xshift=3.0cm] Decision tree:
32 feature combinations
→ UCS score;
(s4) [stage, below of=s3, yshift=-0.4cm, xshift=2.5cm, fill=red!20] Stage 4: LLM
Judge
Fallback; (s4note) [label, right of=s4, xshift=3.0cm] When features are
contradictory/unreliable;
(output_s1)[io,below of = s1dec,yshift = -0.3cm,xshift = 2.5cm,fill =
green!10]DirectUCSScore;(output)[io,below of = s3,yshift = -0.2cm,fill =
blue!10]UCSScore;
[arrow] (input) - (s1); [arrow] (s1) - (s1dec); [arrow] (s1dec) - node[above, font=] Yes
(output,1); [arrow] (s1dec) - node[right, font =] No(s2); [arrow] (s2) - (s3); [arrow] (s3) -
-(output); [darrow,bendright = 30] (s3.east) to node[right, font =
] Lowconfidence(s4.west); [arrow] (s4) - + + (0, -0.8) - (output.south);
[on background layer] [fill=blue!5, rounded corners, fit=(s1)(s1note)(s1dec), inner sep=5pt] ;
[fill=orange!5, rounded corners, fit=(s2)(s2note), inner sep=5pt] ; [fill=green!5, rounded
corners, fit=(s3)(s3note), inner sep=5pt] ; [fill=red!5, rounded corners, fit=(s4)(s4note), inner
sep=5pt] ;

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UCS Engine

four-stage classification pipeline. Easy cases are resolved by pattern matching (Stage 1); the remainder proceed to structured LLM extraction (Stage 2) and deterministic mapping (Stage 3). A fallback LLM judge (Stage 4) handles low-confidence cases.